The Universal Tiling Conjecture for periods at most five

Candidate partial result for arXiv:1209.0666

12 August 2026

Status: candidate substantial partial result, likely valid. The Universal Tiling Conjecture UTC(p) of Dutkay and Jorgensen holds for every $1 \le p \le 5$. Hence every bounded measurable spectral set of measure one with a spectrum of integer period at most five tiles the line. The arbitrary-period conjecture remains open.

1 Source problem

Dutkay and Jorgensen [1] formulate the following conjecture on PDF page 2 (Conjecture 1.4). Let $\Gamma = \{\lambda_0 = 0, \lambda_1, \ldots, \lambda_{p-1}\} \subset \mathbb{R}$ have p elements and at least one spectrum of the form $p^{-1}A$ with $A \subset \mathbb{Z}$. If $A_1, \ldots, A_m \subset \mathbb{Z}$ are any finite family such that $p^{-1}A_i$ is a spectrum for Γ, must there be one set $\mathcal{T} \subset \mathbb{Z}$ for which

$$A_i \oplus \mathcal{T} = \mathbb{Z} \qquad (1 \le i \le m)? \tag{UTC(p)}$$

Here $\oplus$ means that every integer has a unique representation.

where λ_i in $[0, p)$ are some distinct real numbers.

Definition 1.3. Let A be a finite subset of $\mathbb{R}$. We say that A is *spectral* if there exists a finite set Λ in $\mathbb{R}$ such that $\{e_\lambda : \lambda \in \Lambda\}$ is a Hilbert space for $L^2(\delta_A)$ where δ_A is the atomic measure $\delta_A := \sum_{a \in A} \delta_a$ and δ_a is the Dirac measure at a. We call Λ a *spectrum* for A.

We formulate the following "Universal Tiling Conjecture" for a fixed number $p \in \mathbb{N}$:

Conjecture 1.4. **[UTC(p)]** *Let $p \in \mathbb{N}$. Let $\Gamma := \{\lambda_0 = 0, \lambda_1, \ldots, \lambda_{p-1}\}$ be a subset of $\mathbb{R}$ with p elements. Assume Γ has a spectrum of the form $\frac{1}{p}A$ with $A \subset \mathbb{Z}$. Then for every finite family $A_1, A_2, \ldots, A_n$ of subsets of $\mathbb{Z}$ such that $\frac{1}{p}A_i$ is a spectrum for Γ for all i, there exists a common tiling subset $\mathcal{T}$ of $\mathbb{Z}$ such that the set A_i tiles $\mathbb{Z}$ by $\mathcal{T}$ for all $i \in \{1, \ldots, n\}$.*

Figure 1: Conjecture 1.4 in arXiv:1209.0666, source PDF page 2.

2 Idea of the proof

Encode the integer frequency a by the unimodular row vector

$$v(a) = \left(e^{2\pi i a \lambda_j / p}\right)_{j=0}^{p-1} \in \mathbb{T}^p. \tag{1}$$

Then $p^{-1}A$ is a spectrum for Γ exactly when the vectors $p^{-1/2}v(a)$, $a \in A$, are an orthonormal basis of $\mathbb{C}^p$. Moreover, $v(a+b) = v(a)v(b)$ coordinatewise. Thus the problem concerns Hadamard bases cut out of one cyclic character orbit.

For $p = 2, 3$, a zero sum of p unit vectors is rigid: it is an antipodal pair or an equilateral triple. A gcd then puts every A_i on the same complete residue scale. For $p = 4$, every dephased Hadamard matrix has rows $1, u, w, uw$ with $u^2 = 1$; the remaining two-point quotient bases have one fixed 2-adic scale. For $p = 5$, the unique Hadamard equivalence class is the Fourier matrix, so every basis is the unique order-five subgroup of the cyclic orbit. These descriptions exhibit a common complement.

3 The low-period theorem

Theorem 1. *The Universal Tiling Conjecture* $\mathrm{UTC}(p)$ *holds for*

$$p \in \{1, 2, 3, 4, 5\}.$$

More explicitly, for every admissible Γ *and every finite family of integer spectra* $p^{-1}A_i$ *of* Γ*, the proof below constructs one periodic set* $\mathcal{T} \subset \mathbb{Z}$ *such that* $A_i \oplus \mathcal{T} = \mathbb{Z}$ *for every* i*.*

Proof. The case $p = 1$ is immediate. We use the encoding (1) throughout.

The cases $p = 2, 3$**.** Let D be the finite set of all nonzero differences $a - b$ with $a, b \in A_i$ for some i, and put

$$g = \gcd\{|d| : d \in D\}.$$

For each $d \in D$, orthogonality gives

$$\sum_{j=0}^{p-1} e^{2\pi i d\lambda_j/p} = 0. \tag{2}$$

If $p = 2$, (2) says $e^{\pi i d\lambda_1} = -1$, so $d\lambda_1$ is an odd integer. Since g is an integral linear combination of the elements of D, $c = g\lambda_1$ is an integer. It must be odd, and then $(d/g)c = d\lambda_1$ odd implies that d/g is odd for every $d \in D$.

If $p = 3$, the elementary equality

$$1 + z + w = 0, \qquad |z| = |w| = 1,$$

forces $\{z, w\} = \{e^{2\pi i/3}, e^{4\pi i/3}\}$. Therefore $d\lambda_1$ and $d\lambda_2$ are integers not divisible by 3. Again $c = g\lambda_1 \in \mathbb{Z}$, now $3 \nmid c$, and hence $3 \nmid d/g$ for every $d \in D$.

In either case, the elements of each A_i are congruent modulo g, and after subtracting one element and dividing by g, their p residue classes modulo p are pairwise distinct. They are therefore all the classes $0, 1, \ldots, p-1$. It follows directly, or by reducing modulo pg, that the single set

$$\mathcal{T} = \{0, 1, \ldots, g-1\} + pg\mathbb{Z} \tag{3}$$

tiles $\mathbb{Z}$ with every A_i.

The case $p = 4$**.** Choose one integer spectrum $A_0/4$, translate A_0 so that $0 \in A_0$, and form the dephased Hadamard matrix with rows $v(a)$, $a \in A_0$. The classification of 4×4 complex Hadamard matrices says that, after row and column permutations, its rows have the form

$$\mathbf{1}, \quad u, \quad w, \quad uw, \qquad u^2 = \mathbf{1}, \quad u \neq \mathbf{1}; \tag{4}$$

this is the usual one-parameter family $F_4^{(1)}(t)$ recalled in Herr–Wiegand [2]. Since every row lies in the cyclic orbit $v(\mathbb{Z})$, (4) gives a nonzero integer in the kernel of v. Write

$$\ker v = q\mathbb{Z}.$$

The image is cyclic of order q, and its unique element of order two is $v(q/2)$. Also

$$\ell_j := \frac{q\lambda_j}{4} \in \mathbb{Z}, \qquad v(n) = \left(e^{2\pi i n \ell_j/q}\right)_{j=0}^{3}. \tag{5}$$

The residues $L = \{\ell_0, \dots, \ell_3\} \subset \mathbb{Z}/q\mathbb{Z}$ are distinct.

Apply (4) to each A_i after translating one of its elements to zero. Uniqueness of the order-two element in a cyclic group shows that, modulo q, A_i is a translate of

$$J(e_i) = \{0, N, e_i, e_i + N\}, \qquad N = q/2. \tag{6}$$

We next show that all e_i have the same 2-adic valuation. Orthogonality of the rows $v(0)$ and $v(N)$ says that L has two even and two odd elements. Fix the two even representatives ℓ', ℓ'' and put $\delta = \ell' - \ell''$, an even nonzero integer modulo q. For a row set (6), orthogonality of $v(e_i)$ to both $v(0)$ and $v(N)$ gives

$$\sum_{\ell \in L} e^{2\pi i e_i \ell/q} = 0, \qquad \sum_{\ell \in L} (-1)^{\ell} e^{2\pi i e_i \ell/q} = 0.$$

Adding the equations shows that the two even terms cancel, hence

$$e_i \delta \equiv q/2 \pmod{q}. \tag{7}$$

Because δ is even, (7) first implies $4 \mid q$. If $s = \nu_2(q)$, then writing (7) as $e_i\delta = (q/2)(2k+1)$ yields

$$\nu_2(e_i) = s - 1 - \nu_2(\delta) =: t < s - 1, \tag{8}$$

independently of i.

Set $d = \gcd(N, e_1, \dots, e_m)$. Equation (8) says that N/d is even and every e_i/d is odd. In $\mathbb{Z}/N\mathbb{Z}$ define

$$C = \{r + 2dk : 0 \le r < d,\ 0 \le k < N/(2d)\}. \tag{9}$$

Translation by any e_i interchanges C and its complement, so $\{0, e_i\} \oplus C = \mathbb{Z}/N\mathbb{Z}$. Viewing C in $\mathbb{Z}/q\mathbb{Z}$, (6) gives

$$J(e_i) \oplus C = \mathbb{Z}/q\mathbb{Z}$$

for every i. Translates of $J(e_i)$ have the same complement. Consequently $C + q\mathbb{Z}$ is a common tiling set for all the original A_i.

The case $p = 5$. Again start with one translated spectrum and its dephased Hadamard matrix. Haagerup's order-five classification, in the explicit form of Matolcsi–Ruzsa–Weiner [3, Proposition 2.1], says that every 5×5 complex Hadamard matrix is equivalent to the Fourier matrix F_5. After dephasing, its rows are, up to permutations,

$$\mathbf{1}, r, r^2, r^3, r^4, qquad r^5 = \mathbf{1}, quad r \neq \mathbf{1}. \tag{10}$$

Because these rows lie in $v(\mathbb{Z})$, (10) again makes $\ker v = q\mathbb{Z}$ for some q. The cyclic image has an element of order five, so $5 \mid q$, and it has a unique subgroup of order five. Applying the same classification to every A_i shows that, modulo q, each A_i is a translate of

$$\{0, q/5, 2q/5, 3q/5, 4q/5\}.$$

Thus the single set

$$\{0, 1, \dots, q/5 - 1\} + q\mathbb{Z}$$

is a common tiling complement. This completes all five cases. $\square$

4 Consequence for spectral subsets of the line

Corollary 2. *Let $\Omega \subset \mathbb{R}$ be bounded, measurable, spectral, and $|\Omega| = 1$. If Ω has a spectrum*

$$\Lambda = \{\lambda_0 = 0, \ldots, \lambda_{p-1}\} + p\mathbb{Z}$$

with $p \leq 5$, then Ω tiles $\mathbb{R}$ by translations. A tiling set may be chosen inside $p^{-1}\mathbb{Z}$.

Proof. Proposition 2.8 of the source [1] decomposes Ω into p-point integer fibers Ω_x which, for almost every x, have the common spectrum $p^{-1}\{\lambda_0, \ldots, \lambda_{p-1}\}$. Boundedness leaves only finitely many distinct fibers. Theorem 1 gives them a common integer tiling complement. The fiber reconstruction in the same proposition then tiles Ω by the corresponding subset of $p^{-1}\mathbb{Z}$. $\square$

5 Verification, novelty, and limitations

The finite-cyclic audit in `code/verify_low_period.py` enumerated 25,677 primitive column sets over the ranges

$$q \leq 60, 36, 24, 18 \quad \text{for} \quad p = 2, 3, 4, 5,$$

respectively. It found 1,527 nonempty spectrum families containing 1,677 normalized spectra and verified the common complements above in every case. In particular, the $p = 4$ audit includes fixed column sets with more than one row spectrum. This computation is only a bounded check; the proof is the argument above.

The local run indexes and bounded searches through 12 August 2026 used the source title and arXiv id, "Universal Tiling Conjecture" with $p = 2, 3, 4, 5$, low-order Hadamard submatrices, common spectra, and common tiling complements. No statement of UTC(p) for all $p \leq 5$ was found. Herr and Wiegand [2] classify the size-two and size-three Fourier-Hadamard compatibility data and end by calling the relation between a row primitive set and its tiling set a future direction; their paper supplies nearby context, not the common complement theorem proved here. Recent literature still describes the one-dimensional general conjecture as open.

The method has a structural boundary at $p = 6$: zero sums of six unit vectors and order-six complex Hadamard matrices are non-rigid, and a complete order-six classification is unavailable. No claim is made for $p \geq 6$ or for the full one-dimensional Fuglede conjecture. Human review should focus on the dephasing-to-cyclic-subgroup step in the $p = 4, 5$ cases and on the common quotient complement (9).

References

[1] D. E. Dutkay and P. E. T. Jorgensen, *On the universal tiling conjecture in dimension one*, J. Fourier Anal. Appl. 19 (2013), 467–477; arXiv:1209.0666, doi:10.1007/s00041-013-9264-7.

[2] J. E. Herr and T. M. Wiegand, *Identifying complex Hadamard submatrices of the Fourier matrices via primitive sets*, Linear Algebra Appl. 617 (2021), 121–150; arXiv:1909.13145, doi:10.1016/j.laa.2021.01.017.

[3] M. Matolcsi, I. Z. Ruzsa, and M. Weiner, *Real and complex unbiased Hadamard matrices*, arXiv:1201.0631, 2012.